

STAT 482 - Capstone in Analytics for Business

Building an Interactive Business Analytics Dashboard

Using ChatGPT and Excel Data

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Case Study 7 | 180 transactions | R Shiny

The Brief

What the client asked for, who uses it, and what “good” looks like

01

Client

Mid-sized regional retailer wants managers and analysts to inspect mixed-type Excel data without writing code.

02

Goal

Upload a workbook, see descriptive stats, frequency tables, and grouped summaries on demand - and trust the numbers.

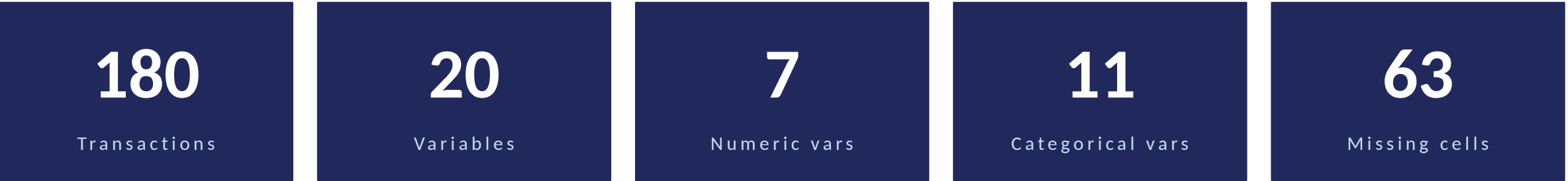
03

Constraint

Use ChatGPT as a co-pilot, but verify every statistic, every line of code, and every interpretation before shipping.

Data at a Glance

STAT482_sample_dashboard_dataset.xlsx - Retail_CX_Data sheet



What's in the workbook

- Quantitative: Order_Value, Discount_Rate, Wait_Time, Delivery_Days, Items, Satisfaction (1-5), NPS (-100 to 100)
- Qualitative: Region, Channel, Product_Category, Customer_Segment, Loyalty_Tier, Payment_Method, Complaint_Theme
- Binary flags: Complaint_Flag, Return_Flag
- Identifier / date / free text: Transaction_ID, Order_Date, Manager_Notes

Missing-value pattern

Column	Missing	Treatment
Wait_Time_Min	54 (30%)	Structural - online has none
Complaint_Theme	133 (74%)	Tied to Complaint_Flag = No
Loyalty_Tier	3 (1.7%)	Kept as explicit (missing)
Manager_Notes	6 (3.3%)	Free text, flagged only

Dashboard Architecture

Sidebar controls + 7-tab main panel, built on Shiny + readxl + dplyr + ggplot2 + DT

S I D E B A R

- Excel upload
- Worksheet selector
- Auto header-row detect
- Filters - Region, Channel, Category, Segment, Tier
- Type-detection helper

Overview

KPIs, type table, missing-values plot

Numeric Summary

n, missing, mean, median, SD, Q1-Q3, min, max

Categorical Summary

Frequency table for any qualitative var

Grouped Summary

Numeric variable summarised by category

Plot

Histogram or bar chart, optional grouping

Raw Data

Searchable, column-filterable preview

About

Purpose, features, ChatGPT-use disclosure

Headline Numbers

Descriptive snapshot of the seven numeric variables (n = 180)

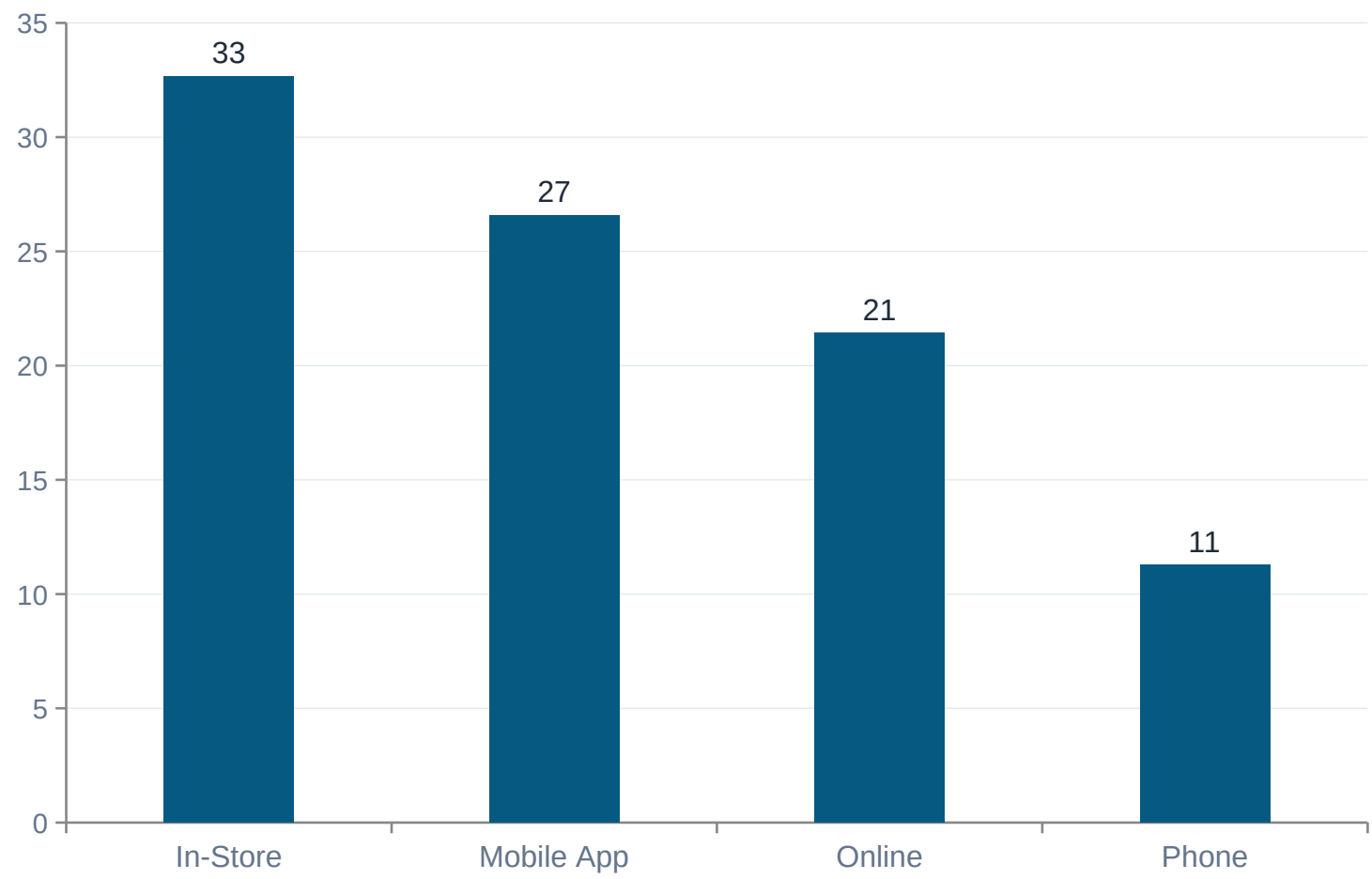
Variable	N	Missing	Mean	Median	SD	Min	Max
Items_Purchased	180	0	4.38	4.00	1.78	1	11
Order_Value_USD	180	0	116.71	117.48	47.44	16.50	250.98
Discount_Rate	180	0	0.107	0.103	0.056	0.000	0.262
Wait_Time_Min	126	54	6.66	6.95	3.67	1.0	14.5
Delivery_Days	180	0	2.12	2.00	1.96	0	7
Satisfaction_Score	180	0	3.46	4.00	1.14	1	5
NPS_Score	180	0	25.07	38.00	40.35	-75	87

What this says

Order value averages USD 117 with wide spread (SD ~ 47). NPS mean 25 but median 38 - long negative tail. Satisfaction skews positive (median 4) but ~ 1/3 of customers sit at 3 or below.

Channel Story: Phone Drags NPS Down

Mean Net Promoter Score by purchase channel



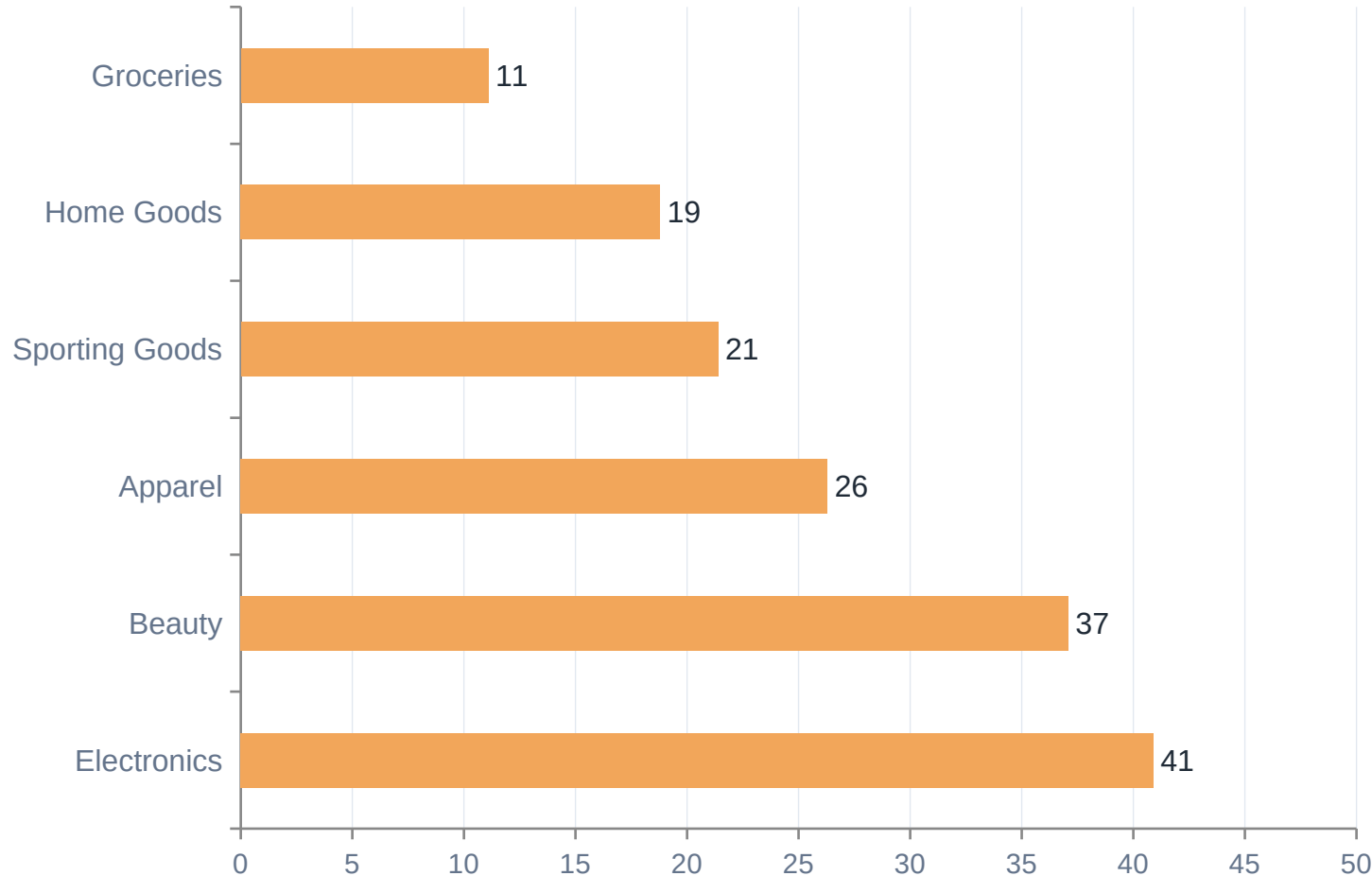
21-pt gap
In-Store vs. Phone NPS

n = 25
Phone sample is small - watch noise

MANAGER ACTION
Sample 30 phone calls and check training, queue length, and escalation handling.

Where Complaints Concentrate

Complaint rate (%) by product category - portfolio average is 26%



Read this as

- Electronics and Beauty are far above the 26% portfolio rate.
- Groceries is unusually clean - a useful counter-example.
- Per-category n's are small (16-42) so treat the gap, not the exact rate, as the signal.
- Action: pair next 30 days of complaints with returns data to separate defects from expectation mismatch.

Validation: Trust, Then Verify

Every dashboard number was reproduced by a second method before this report

1

Excel pivot

Recreated Region x Order_Value and Channel x NPS as Excel pivots. Match to two decimal places.

2

Independent R summary

Re-ran summary() in base R on the numeric block. Identical n, mean, median, SD, min, max.

3

Edge cases

Filter to zero-row subsets (Southeast x Phone). App returns N = 0 cleanly, no crash.

4

Missing-value audit

na.rm = TRUE applied consistently. Frequency tables show NAs as an explicit category.

Working with ChatGPT

Used as a junior collaborator - never as the analyst of record

HELPED

- Brainstormed the 7-tab Shiny layout before any code was written.
- Drafted the auto header-row detector after we described the title-row problem.
- Suggested showing medians alongside means for grouped summaries on small subgroups.
- Speeded up boilerplate around DataTable rendering and DT options.

REJECTED / CORRECTED

- Mean-imputation of Wait_Time_Min - we recognised the missingness as structural, not random.
- A regression-based “loyalty score” - out of scope for a descriptive dashboard, would have given false rigour.
- table() without useNA = “ifany” - silently dropped 3 missing Loyalty_Tier rows. Caught and fixed.
- Discipline: every stat re-computed in Excel or base R; every helper function read line-by-line.

Recommendations

Five concrete moves the manager can take this quarter

01

Fix the Phone channel

NPS gap of 21 points vs. In-Store. Sample calls and review queue / escalation.

02

Audit Electronics & Beauty

Complaint rates ~40% vs. 26% portfolio average. Pair with returns data for 30 days.

03

Backfill Loyalty_Tier

3 unlabelled rows = small but real CRM data-entry gap. One-time review of recent sign-ups.

04

Add a time-series tab

Once 6-12 months of Order_Date are available, watch for drift in NPS and complaint mix.

05

Adopt for weekly review

Light enough for one Shiny Server. 15-min weekly check with regional analyst keeps it useful.

Thank you - questions welcome.